18 (a) Express $\sqrt{3} + \sqrt{12}$ in the form $a\sqrt{3}$ where a is an integer.

(a) √12 = √(4 × 3) = 2√3

1 mark
Extract square factor 4.

√3 + 2√3 = 3√3

1 mark
3√3.

3√3

(2)

(b) Express $\left(\frac{1}{\sqrt{3}}\right)^7$ in the form $\frac{\sqrt{b}}{c}$ where b and c are integers.

(b) $(1/\sqrt{3})^7 = 1/(27\sqrt{3})$

1 mark
Simplify the seventh power.

Rationalise: $1/(27\sqrt{3}) \times \sqrt{3}/\sqrt{3}$

1 mark
Multiply by √3/√3.

$= \sqrt{3}/81$

1 mark
√3/81.

$$\frac{\sqrt{3}}{81}$$

(3)

(Total for Question 18 is 5 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA